PIEZO INSULIN PUMP v2 — REV B DIGITAL HANDOVER PROTOTYPE
BENCH-ONLY: water or 0.9 % saline. NEVER connect to a person or an animal. No real insulin.
NOT FABRICATION-APPROVED. NOT VALIDATED. Contains TBD and PLACEHOLDER parts (Status field / BOM).
Clean ERC/DRC does not prove electrical correctness. Rev A bench validation has not been performed.
Design basis: handoff documents 10 Oct 2026; locked decisions K-1..K-18 preserved. See docs/handover/.

H901
M3

H902
M3

H903
M3

H904
M3

Power input & charger

Battery input, fuse, master switch, BQ25188 charger

File: power_input.kicad_sch

3V3 buck

TPS62840 3.3 V logic rail

File: power_3v3.kicad_sch

48V flyback

LM51551 non-isolated flyback, 48 V motor rail

File: power_48v.kicad_sch

Kill path (fail-off)

48 V enable kill path: main MCU enable, supervisor veto

File: safety.kicad_sch

Main MCU STM32WBA55

Main MCU STM32WBA55CG: dosing, BLE, encoder, PMD401 SPI

File: mcu_main.kicad_sch

Supervisor STM32C031

Supervisor STM32C031C6: heartbeat, park line, 48 V veto, own encoder count

File: mcu_supervisor.kicad_sch

PMD401 & encoder interface

PMD401 servo/status/power/RS-485 connectors, LL06 encoder, protection

File: pmd_interface.kicad_sch

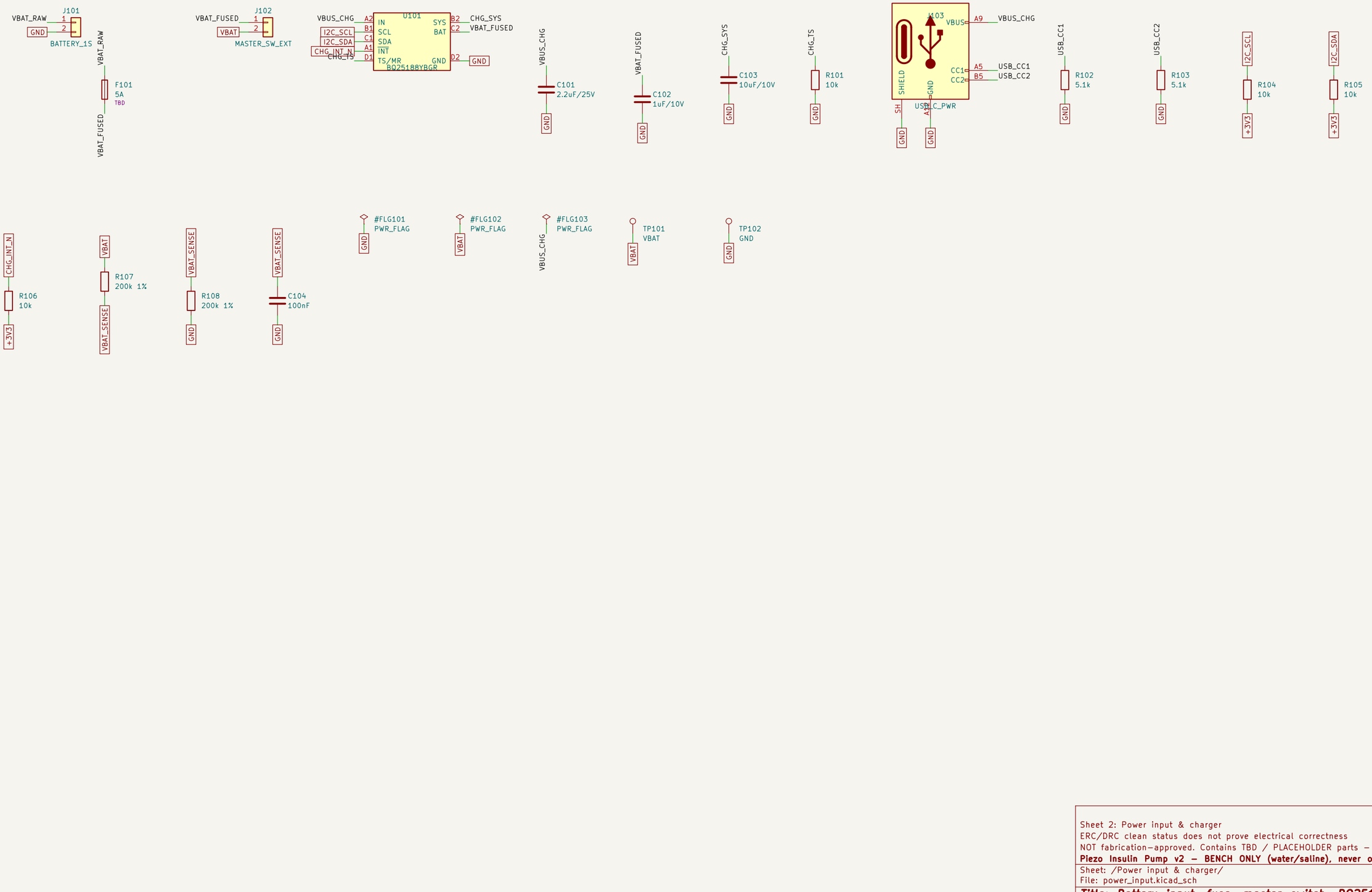
Root: hierarchy, mounting, disclaimers
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts — see Status field and docs/handove
Piezo Insulin Pump v2 — BENCH ONLY (water/saline), never on a person or animal

Sheet: /
File: pump_v2.kicad_sch

Title: Piezo Insulin Pump v2 — Rev B handover prototype

Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 1/8

Battery path (E-01 DECIDED): 1S protected cell -> J101 -> F101 5 A fuse -> J102 master switch -> VBAT.
Charger (K-12 BQ25188) charges the cell on VBAT_FUSED. E-09: the charger is NEVER the load path:
BQ25188 SYS (CHG_SYS) has only its required 10 uF and no load. Charge current set over I2C (OPEN B.19).
Cell protection is inside the pack (K-12 '1S protected cell'). Cell size/connector OPEN B.20.



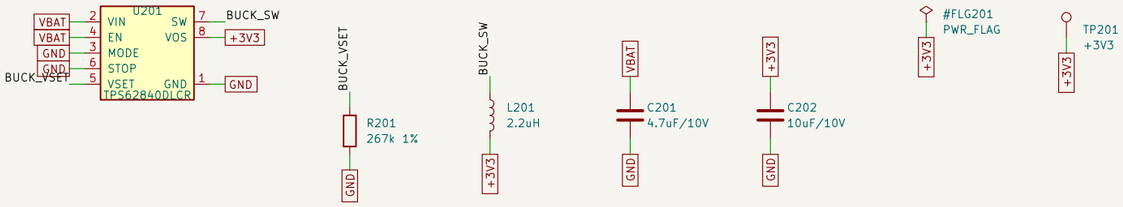
Sheet 2: Power input & charger
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts — see Status field and docs/handover
Piezo Insulin Pump v2 — BENCH ONLY (water/saline), never on a person or animal

Sheet: /Power input & charger/
File: power_input.kicad_sch

Title: Battery input, fuse, master switch, BQ25188 charger

Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 2/8

K-12: TPS62840 for 3.3 V on Rev B. Taken from VBAT (not from the 48 V chain), as decided for Rev A.
Below about 3.4 V in, the buck runs in 100 % mode and 3V3 follows VBAT (E-02 >=3.0 V while dosing; S-13 3.4 V stop).
Note CF-8: E-02 wording says 'LDO' (Rev A); Rev B uses the decided TPS62840 buck.



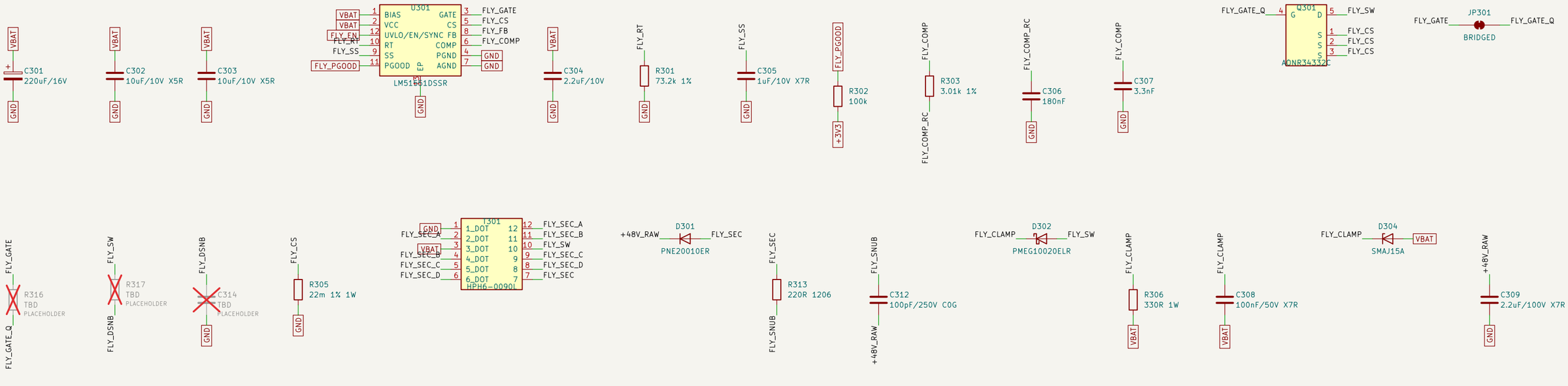
Sheet 3: 3V3 buck
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts — see Status field and docs/handove
Piezo Insulin Pump v2 — BENCH ONLY (water/saline), never on a person or animal

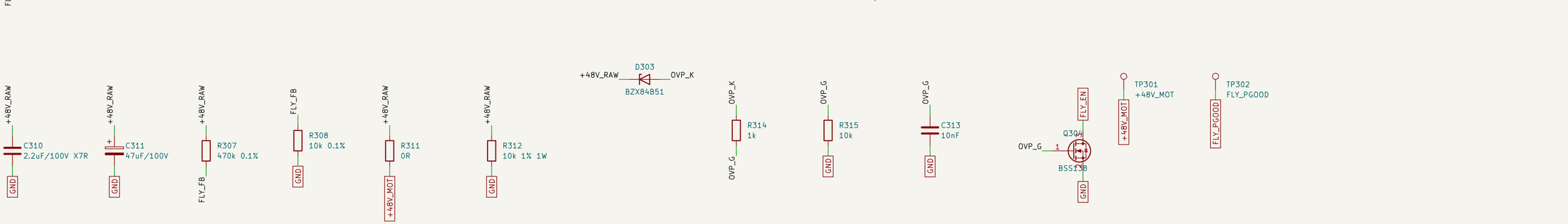
Sheet: /3V3 buck/
File: power_3v3.kicad_sch

Title: TPS62840 3.3 V logic rail

Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 3/8

K-12 / DECIDED 29 Sep: Rev B 48 V stage is an LM51551 flyback (plain boost hits the 93 % duty limit).
Topology follows TI SNVSB75E Figure 10-12 'Typical Non-Isolated Flyback'; BIAS tied to VCC for 1S start (>=2.97 V).
Design calculations: docs/handover/FLYBACK_DESIGN.md (fsw ~300 kHz, n = Ns/Np = 5, Lp ~7 uH, Ipk ~3.1 A at 3.0 V / 100 mA).
T301 = Coilcraft HPH6-0090L: pri 3-10, sec = 5 windings in series. Leakage is NOT published: clamp and snubber values need bench tuning.
Compensation values are PRELIMINARY (depend on T301 and the PMD401 cable/bulk capacitance) - simulate/bench before use.
Kill path: UVLO/EN driven by FLY_EN (sheet 'Kill path'). EN < 0.4 V min (SNVSB75E) = shutdown. A flyback has no DC path
from VBAT to the output, so EN low de-energises the rail. EC01 (owner approved architecture): no output switch; rail control is EN; R312 bleeder and D303 clamp fitted.
E-06: every 48 V capacitor rated >= 100 V. E-03: 45.6-50.4 V at the PMD401.





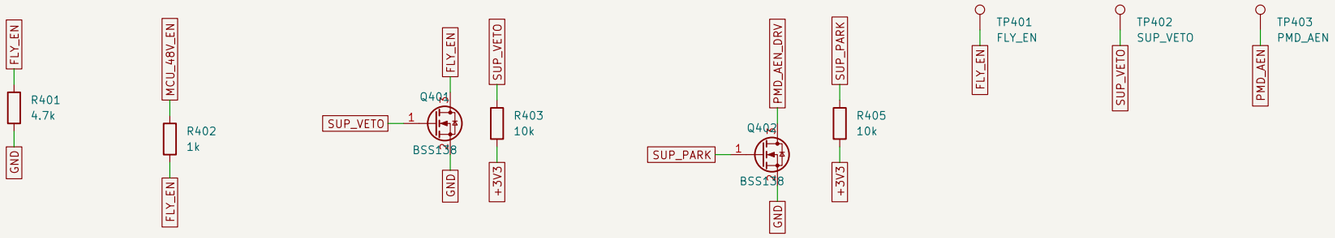
Sheet 4: 48V flyback
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts - see Status field and docs/handover
Piezo Insulin Pump v2 - BENCH ONLY (water/saline), never on a person or animal

Sheet: /48V flyback/
File: power_48v.kicad_sch

Title: LM51551 non-isolated flyback, 48 V motor rail

Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 4/8

K-10 DECIDED (fail-off): 4.7 k pull-down on the enable, main MCU drives it through 1 k, supervisor veto via one BSS138.
On Rev B the enable node is LM51551 UVLO/EN (FLY_EN). EN rising threshold 0.4-0.7 V, UVLO 1.425-1.575 V (SNVSB75E).
MCU high: 3.3 V x 4.7/(4.7+1) = 2.72 V -> run. Pull-down only: 0 V -> shutdown. BSS138 on: ~0 V -> shutdown.
Veto default (owner decision 10 Oct 2026, OPEN item 3 closed): R403 10k PULL-UP fitted -> veto ON whenever the supervisor
is in reset, crashed with its pin high/high-Z, or not running; firmware must drive PB0 LOW to permit 48 V. (R404 removed.)
PMD401 park (owner decision 10 Oct 2026, OPEN item 4): open-drain Q402 BSS138 with R405 10k gate PULL-UP -> AEN held LOW
(parked) by default; the PMD401's own 22k pull-up provides the run level, so no 3.3/5 V level question. Supervisor drives
PB1 (SUP_PARK) LOW to unpark. BSS138 VGS(th) 0.9-1.5 V (Nexperia BSS138P datasheet). SG-03/SG-05 still required on hardware.



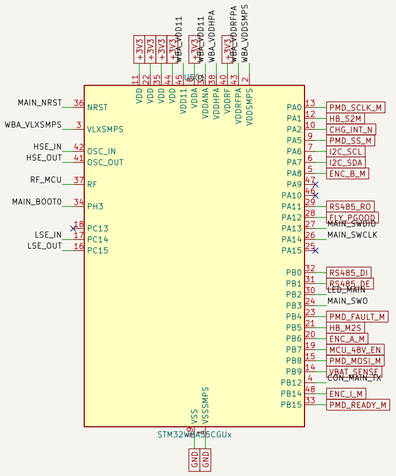
Sheet 5: Kill path (fail-off)
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts - see Status field and docs/handover
Piezo Insulin Pump v2 - BENCH ONLY (water/saline), never on a person or animal

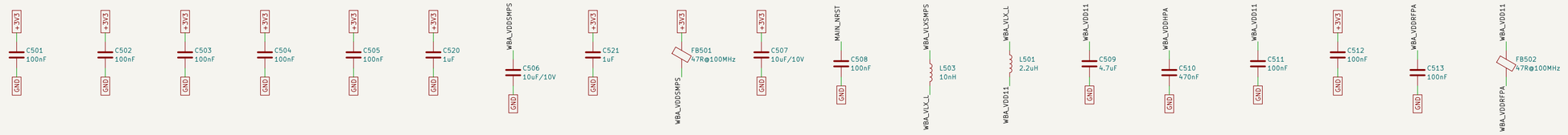
Sheet: /Kill path (fail-off)/
File: safety.kicad_sch

Title: 48 V enable kill path: main MCU enable, supervisor veto

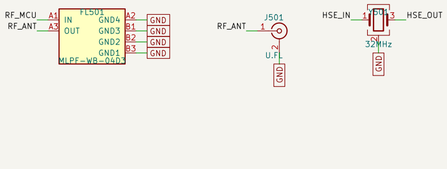
Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 5/8

Pin map DERIVED from ST pin data in the KiCad STM32WBA55CGUx symbol: CONFIRM IN STM32CubeMX before layout freeze.
Encoder on TIM2_CH1/CH2 (PB6/PA8) — TIM2 width must be confirmed in RM0493 (CF-21/FW-08).
SPI3 -> PMD401 (mode 3, <= 500 kbit/s, PMD401 manual 150025-01 sec.5.2). USART2 + DE -> MAX3485. LPUART1 <-> supervisor link.
USART1 TX-only console (PA8, the only USART1_RX pin, carries the encoder).
ECO1: supplies per DS14127 Figure 15 (SMPS scheme), AN6373 and MB1803 B-01. VDDRFPA on the SMPS output limits TX to about +3 dBm (AN6373 sec.7);
for +10 dBm fit R513 and remove FB502. RF: MLPF-WB-0403 with a 63 ohm line from pin 37 and a 50 ohm line to J501. HSE and LSE per AN5948 Table 5.









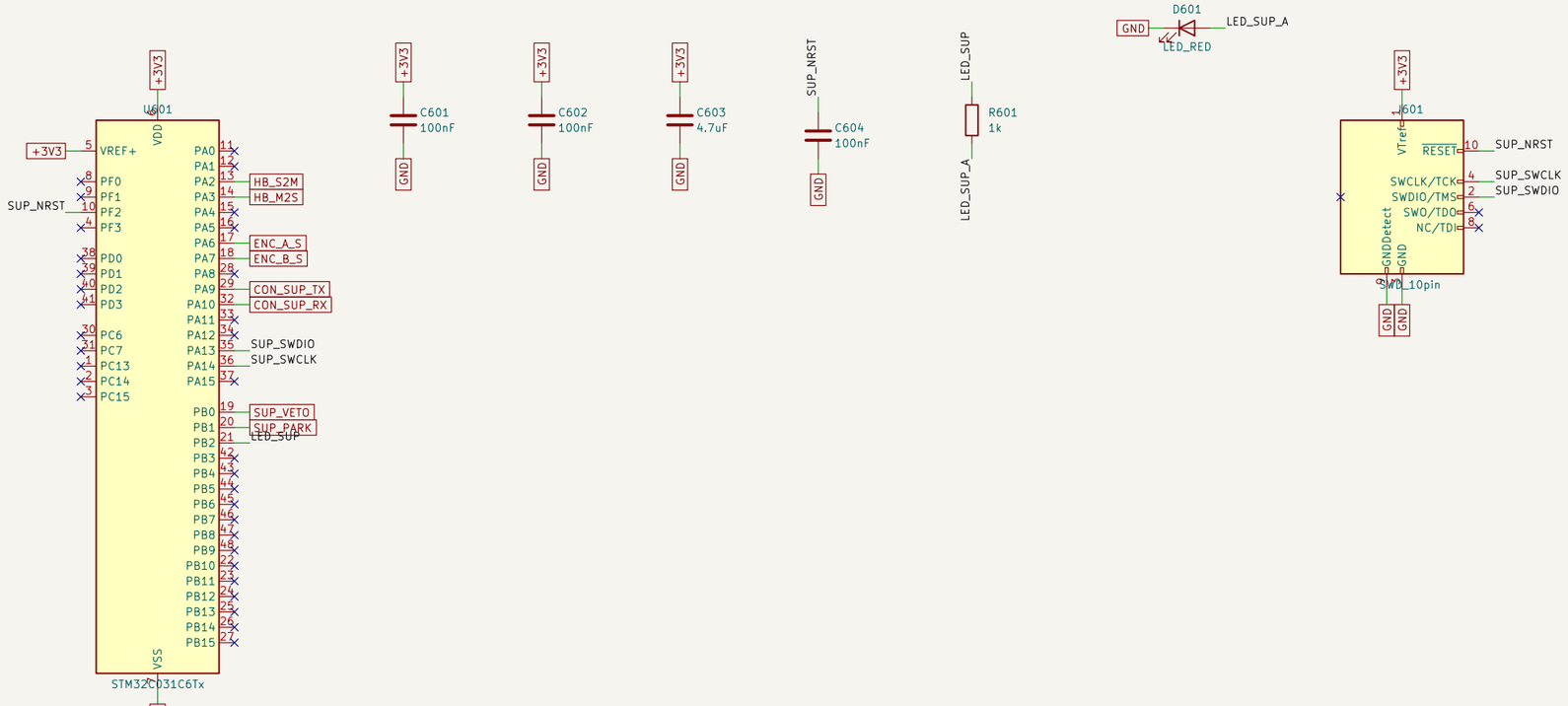
Sheet 6: Main MCU STM32WBA55
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts — see Status field and docs/handover
Piezo Insulin Pump v2 — BENCH ONLY (water/saline), never on a person or animal

Sheet: /Main MCU STM32WBA55/
File: mcu_main.kicad_sch

Title: Main MCU STM32WBA55CG: dosing, BLE, encoder, PMD401 SPI

Size: A2	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 6/8

K-6 / F-09: separate MCU, separate code base. It can only STOP: veto on the 48 V enable (SUP_VETO) and park line (PMD_AEN).
S-07 (approved 10 Oct): encoder A/B also into TIM3 CH1/CH2 via its own series resistors.
Pin map DERIVED from the KiCad STM32C031C6Tx symbol (ST pin data): CONFIRM in CubeMX. NRST assumed on PF2 (PF2-NRST) - VERIFY.
Package LQFP48 (C6T6) ASSUMED; the C031C6 also exists in UFQFPN48 (C6U6).



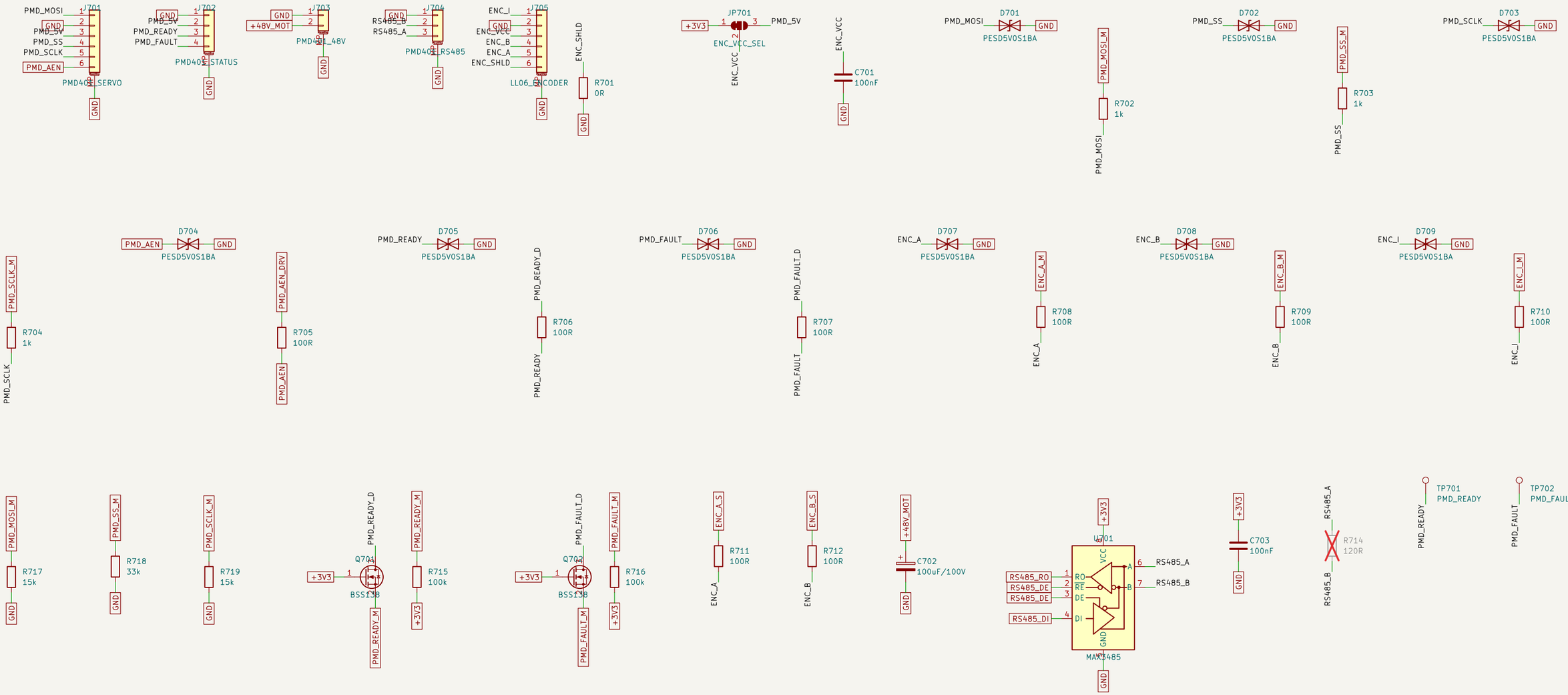
Sheet 7: Supervisor STM32C031
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts - see Status field and docs/handove
Piezo Insulin Pump v2 - BENCH ONLY (water/saline), never on a person or animal

Sheet: /Supervisor STM32C031/
File: mcu_supervisor.kicad_sch

Title: Supervisor STM32C031C6: heartbeat, park line, 48 V veto, own en

Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 7/8

Connector pinouts CONFIRMED from PiezoMotor PMD401 Technical Manual 150025-01 sec.3.4 / 3.5 (servo mode). All JST SH 1.0 mm.
Servo connector = the PMD401 6-pole sensor connector in servo mode; the LL06 encoder therefore connects to THIS board (J705).
J705 mirrors the PMD401 sensor-connector pinout so the kit encoder cable can plug in - VERIFY against the delivered cable.
E-07: TVS + 100 R on every PMD401 signal line. PMD401 inputs have internal pull-ups (10k/~20k/22k) to an UNSTATED rail (IF-02).
ECO1: READY and FAULT reach the MCU through BSS138 level shifters Q701/Q702: the MCU pin never sees above 3.3 V whatever the PMD401 rail, in reset or unpowered.
FIRMWARE: READY and FAULT both read LOW when the PMD401 is unpowered, so trust them only after FLY_PGOOD is high.
AEN has a 22k internal pull-up (manual); it is driven open-drain by Q402 (sheet Kill path): parked unless the supervisor unparks.
Encoder supply (OPEN item 12): JP701 default 3V3 (logic rail, survives 48 V switching); alternative PMD401 5V Out.



Sheet 8: PMD401 & encoder interface
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts - see Status field and docs/handove
Piezo Insulin Pump v2 - BENCH ONLY (water/saline), never on a person or animal
Sheet: /PMD401 & encoder interface/
File: pmd_interface.kicad_sch
Title: PMD401 servo/status/power/RS-485 connectors, LL06 encoder, pro
Size: A3 | Date: 2026-10-10 | **Rev: B-draft-handover**
KiCad E.D.A. 10.0.3 | Id: 8/8